

A Hybrid Transformer-Based Sentiment Analysis Model with Emoji-Slang Normalization and Sarcasm Detection

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Abstract:

Social media has become a major source of real-time opinions, emotions, and public reactions, especially through platforms like X. This is accompanied by a rapid growth in volumes of uninstructed textual data generated by users, creating new opportunities for automated sentiment interpretation. This work outlines a hybrid sentiment analysis framework that merges classic machine learning with modern deep learning for more reliable context-sensitive sentiment classification. This work proposes a system that integrates a transformer-based DistillBERT model for contextual understanding, logistic regression with TF-IDF for lexical precision, and a bidirectional LSTM for sequential dependency learning, all combined through a weighted ensemble approach. An extra preprocessing layer was added to normalize emojis and slang expression, while a sarcasm detection module using RoBERTa adjusts the prediction when irony is detected. This model is trained and evaluated on the Sentiment140 dataset, achieving improved accuracy and robustness compared to typical TF-IDF -based classifiers. These results prove that a hybrid design will provide a greater capacity to handle diverse linguistic expressions, improve performance on noisy data, and give stable predictions across different contexts. The proposed framework has great potential in practical applications such as brand reputation monitoring, social media analytics, and public sentiment tracking. Besides, by incorporating classic interpretability with deep contextual reasoning the system achieves a good balance between transparency and intelligence, thus laying the foundation for future research in emotion-aware and explainable AI- powered sentiment analysis.

Keywords: Sentimental Analysis, Machine learning, Natural Language processing(NLP), Transformer models, DistillBERT, BiLSTM, TF-IDF, Sarcasm detection, Social media analytics